Olivier Juan

Research Engineer Expert

olivier.juan@edf.fr · olivier.juan@gmail.com · Palaiseau, France

olivierjuan.github.io · olivierjuan · olivier-juan-732a07ba

Google Scholar · 0000-0003-3445-4847

Experience

Jan 2008 – present **EDF R&D · OSIRIS, Palaiseau, France**

- Led the ground-up redesign and production deployment of **Apogène**, EDF's short-term unit commitment software (MILP, branch-and-bound, ADMM, RCSPP, Min Cost Flow). 4 versions deployed in production (V2 2018, V3 2020, V4 2021, V5 2025), saving several tens of millions of euros per major release.
- Managed a team of 10–20 engineers and researchers across the full lifecycle: mathematical modeling, algorithm design, C++ implementation, CI/CD (GitLab), HPC deployment (Slurm, Singularity).
- Integrated ML-based branching heuristics (RL, GNN) into the solver loop; results published at NeurIPS 2025 and AAAI 2026.
- Designed and built the optimization platform for DREEV (EDF × Nuvve joint venture): EV fleet dispatch across NEBEF, Spot, Intraday, and FCR markets (MILP, Lagrangian decomposition). DREEV became first French EV aggregator accredited for FCR by RTE (2022). Ongoing research on Frank-Wolfe decomposition methods as a replacement for Lagrangian decomposition in legacy algorithms.
- Earlier work (2008–2012): task scheduling for maintenance planning, LNG terminal planning & scheduling, combustion and emission optimization for coal power plants.

Jan 2007 – Jan 2008 **École Centrale de Paris, France**

- Combinatorial algorithms for medical imaging, with Nikos Paragios.

Jan 2006 – Jan 2007 **University of Western Ontario, Canada**

- Graph cuts and combinatorial algorithms for computer vision, with Yuri Boykov.

Education

Jan 2006 **École Nationale des Ponts et Chaussées (ENPC), France**, Computer Vision

- Distinction. Dissertation: *On Some Extensions of Level Sets and Graph Cuts & Their Applications to Image and Video Segmentation*

Jan 2002 **ENS Paris-Saclay (formerly ENS de Cachan), France**, Computer Vision

Awards

Jan 2022 **IndusRO'2022 Prize**

Awarded for the industrial impact of Apogène on EDF's power plant operations.
ROADEF (French Society for Operations Research and Decision Aid)

Teaching

2023, 2024 **From Long-term to Short-term Unit Commitment at EDF**

Co-architect and lecturer (courses + practical sessions) for a 5-day master-level intensive program on electricity system planning. Module covers unit commitment optimization across all EDF horizons (weekly, daily, intraday), from MILP modeling to production deployment.
Olivier Beaude, Olivier Juan

Jan 2020 **Game Theory Applied to Electricity System Problems**

Guest lecture on game-theoretic models for electricity markets and smart grid problems, within the master-level course on game theory for smart cities.
Olivier Beaude, Olivier Juan

Supervision

Jan 2023 **Paul Strang — CIFRE PhD**

Co-supervision of a CIFRE industrial PhD on reinforcement learning and model-based planning for branching strategies in branch-and-bound solvers (BBMDP @ NeurIPS 2025, PlanB&B @ AAAI 2026). Expected graduation: 2026.
Olivier Juan, Come Bissuel, Emmanuel Rachelson, Safia Kedad-Sidhoum, Zacharie Ales

Jan 2018 **Marc Etheve — CIFRE PhD**

Co-supervision of a CIFRE industrial PhD on reinforcement learning for variable selection in branch-and-bound (FMSTS @ CPAIOR 2020). Graduated 2021.
Come Bissuel, Olivier Juan, Safia Kedad-Sidhoum, Zacharie Ales

Talks

Oct 2024 **Apprentissage par renforcement pour l'apprentissage de stratégies de branchement**

Invited talk on reinforcement learning for branching strategy optimization in branch-and-bound solvers, based on joint work with CNAM and ISAE-SUPAERO at EDF R&D.
Olivier JUAN

Apr 2021 **EV Flexibility for the Electric System: The FCR Case**

Invited talk on smart EV charging optimization and participation of EV fleets in the Frequency Containment Reserve (FCR) market, in the context of the DREEV joint venture (EDF × Nuvve).
Olivier JUAN
olivierjuan.github.io/mal-folio.github.io/assets/pdf/IRT_systemX_EV_flexibility_for_electric_system.pdf

Recent Publications

Jan 2026 **PlanB&B: Planning in Branch-and-Bound via Model-Based RL**

Strang, Alès, Bissuel, Juan, Kedad-Sidhoum, Rachelson.
Paul Strang, Zacharie Alès, Côme Bissuel, Olivier Juan, Safia Kedad-Sidhoum, Emmanuel Rachelson
arxiv.org/abs/2511.09219

Jan 2025 **BBMDP: A Markov Decision Process for Variable Selection in B&B**

Strang, Alès, Bissuel, Juan, Kedad-Sidhoum, Rachelson.
Paul Strang, Zacharie Alès, Côme Bissuel, Olivier Juan, Safia Kedad-Sidhoum, Emmanuel Rachelson
arxiv.org/abs/2510.19348

Jan 2020 **Reinforcement Learning for Variable Selection in a Branch and Bound Algorithm**

Etheve, Alès, Bissuel, Juan, Kedad-Sidhoum.
Marc Etheve, Zacharie Alès, Côme Bissuel, Olivier Juan, Safia Kedad-Sidhoum
doi.org/10.1007/978-3-030-58942-4_12

Skills

Optimization: MILP, Branch-and-bound, Cutting planes, RCSPP, ADMM, Min Cost Flow, Constraint programming, Decomposition methods, Heuristics, Genetic Algorithm, Bundle methods, Frank-Wolfe decomposition

Solvers: CPLEX, GUROBI, SCIP, FICO Xpress, COIN-OR, HiGHS, Interior Point Methods

Machine Learning: Reinforcement learning, Graph Neural Networks, ML for combinatorial optimization

Frameworks: PyTorch, PyG, RLLib, Ray

Programming Languages: C++, Python, Julia

Spoken Languages: French (native), English (fluent)

Engineering: GitLab CI/CD, Docker, TBB, HPC / Slurm, Singularity, CMake, Conan